

SQUAMOUS CARCINOMA

CELL

AT A GLANCE

With **basal cell carcinoma (BCC)**, **squamous cell carcinoma (SCC)** represents one of the two major types of non-melanoma skin cancers and is among the most common human malignancies. While BCC is more frequent, SCC carries a higher risk of metastasis and accounts for the majority of non-melanoma skin cancer-related deaths.

SCC typically arises from precursor lesions such as **actinic keratosis (AK)** and **Bowen disease (SCC in situ)**, in contrast to BCC, which usually develops de novo. This chapter focuses on the clinical features of invasive cutaneous SCC, which spans a broad spectrum—from localized, treatable tumors to aggressive, deeply infiltrative cancers capable of metastasis and causing death. Clinical presentation varies, although typical lesions are often recognizable. Molecular and cellular mechanisms of SCC carcinogenesis are discussed elsewhere (see Chap. 112).

EPIDEMIOLOGY

Incidence

The exact incidence of basal cell and squamous cell carcinomas is unknown, as these cancers are not uniformly reported to national or state cancer registries, including those maintained by the National Cancer Institute. Nevertheless, over 1 million new non-melanoma skin cancer cases are diagnosed annually in the United States, with approximately 200,000 being SCC.

Although less common than BCC, SCC has a greater potential for metastasis and is responsible for most of the several thousand annual deaths attributed to

non-melanoma skin cancer. In comparison, cutaneous melanoma accounts for about 60,000 cases and approximately 9,000 deaths per year.

Similar increases in SCC incidence have been observed in Australia and the Caribbean.

SCC incidence rises sharply after age 40 and is strongly associated with advanced age. Currently, the lifetime risk of developing SCC among white individuals in the U.S. is about 15%, nearly double what it was two decades ago. Contributing factors likely include increased ultraviolet (UV) radiation exposure (from tanning beds, outdoor activities, clothing trends, and ozone depletion) and increased life expectancy. This upward trend is expected to continue due to ongoing ozone layer depletion and population aging.

Despite rising incidence, mortality from SCC has decreased by approximately 20% over recent decades, likely due to heightened public awareness and more aggressive management of high-risk tumors.

After an initial SCC diagnosis, patients face a 44% to 50% cumulative risk of developing another non-melanoma skin cancer within 3 to 5 years, including an 18% to 30% risk of a second SCC. Additionally, these individuals are at increased risk for developing non-cutaneous malignancies.

Demographics

SCC occurs approximately twice as frequently in men as in women, likely due to greater cumulative UV exposure. Differences in hairstyle and the use of lip protectants (e.g., lipstick with UV protection) may contribute to lower rates of SCC on the ears and lips in women.

There is a strong inverse correlation between skin pigmentation and SCC risk, primarily due to the photoprotective properties of eumelanin. Individuals with fair skin, blue eyes, red or blond hair, poor tanning ability, and Celtic ancestry are at highest risk.

Compared to white individuals, the incidence of SCC is about 30 times lower in African Americans, and the BCC-to-SCC ratio shifts from >4:1 in whites to

0.8:1 in this population. In Tanzanian individuals with albinism, SCC develops universally by early adulthood, and the BCC-to-SCC ratio is only 0.2:1.

Asians and Polynesians, who have intermediate levels of skin pigmentation, show correspondingly intermediate SCC rates.

The **MC1R gene**, which encodes the melanocortin 1 receptor and plays a key role in melanin production, is a major determinant of skin and hair pigmentation. Over 20 polymorphic variants of MC1R have been identified, and certain alleles are associated with an increased risk of SCC independent of phenotypic traits such as skin type or hair color.